

NUTRITIONAL REQUIREMENTS REFERENCE

Inherited Metabolic Diseases

By Age, Gender & Disease Severity

Source: Nutrition Management of Inherited Metabolic Diseases, 2nd Edition (2022)

Editors: Laurie E. Bernstein, Fran Rohr, Sandy van Calcar — Springer Nature Switzerland

All tables are extracted directly from the book guidelines. Always refer to original source for clinical decisions.

#	Disease	Enzyme	Key Restriction	Treatment Goal
1	Phenylketonuria (PKU)	PAH	Phenylalanine	Phe: 120–360 µmol/L
2	Hereditary Tyrosinemia T1	FAH	Phe + Tyrosine	Tyr: 200–600 µmol/L NTBC
3	Homocystinuria (HCU)	CBS	Methionine	tHcy <50–100 µmol/L
4	Maple Syrup Urine Dis. (MSUD)	BCKADH	BCAA (Leu, Ile, Val)	Leu: 100–300 µmol/L
5	Urea Cycle Disorders (UCD)	NAGS/CPS/OTC/ASS/AS	Intact protein	Ammonia <35 µmol/L
6	Glutaric Acidemia T1 (GA-1)	GCDH	Lysine	Lys: low-normal Carnitine normal
7	Propionic/Methylmalonic Acidemia	PNPCC MMA:MCM	Propiogenic AAs	Plasma AA: normal
8	LC-FAOD (VLCAD/LCHAD)	VLCAD / LCHAD	Long-chain fat + fasting	Glucose normal Carnitine normal
9	MCAD Deficiency	MCAD	Fasting only	Glucose normal Carnitine normal
10	Classic Galactosemia	GALT	Galactose / Lactose	RBC Gal-1-P <1 mg/dL
11	GSD Type I (Von Gierke)	G6Pase (Ia) / G6P transporter	Fructose + galactose + fasting	Glucose: 70–120 mg/dL
12	GSD Type III (Cori/Forbes)	AGL (debranching enzyme)	Simple sugars + fasting	Glucose normal CK normal

1. Phenylketonuria (PKU)

Deficient Enzyme Phenylalanine hydroxylase (PAH) Phe → Tyr	Toxic Metabolite Phenylalanine (Phe)	Treatment Goal Blood Phe: 120–360 µmol/L (2–6 mg/dL) Tyrosine: within normal range
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Note: Gender: No difference in requirements except during pregnancy (see Chapter 12). Conversion: 50 mg Phe ≈ 1 g protein.

Daily Nutrient Requirements by Age

Age	Protein (g/kg/d)	Phe (mg/kg/d)	Phe (mg/day)	Tyrosine (mg/day)
Birth–3 mo	2.5–3.0	25–70	130–430	1100–1300
3–<6 mo	2.0–3.0	20–45	135–400	1400–2100
6–<9 mo	2.0–2.5	15–35	145–370	2500–3000
9–<12 mo	2.0–2.5	10–35	135–330	2500–3000
1–4 yr	1.5–2.0	—	200–320	2800–3500
4 yr–Adult	120–140% DRI	—	200–1100	4000–6000

Initial Phe Prescription After Washout

Diagnostic Blood Phe	Starting Phe (mg/kg)
<600 µmol/L (<10 mg/dL)	70
600–1200 µmol/L (10–20)	55
1200–1800 µmol/L (20–30)	45
1800–2400 µmol/L (30–40)	35

Reference: Table 10.3 | SERN-GMDI PKU Guidelines | van Calcar S. Springer 2022

2. Hereditary Tyrosinemia Type 1 (HT-1)

Deficient Enzyme Fumarylacetoacetate hydrolase (FAH)	Toxic Metabolite Succinylacetone + succinylacetoacetate	Treatment Goals Phe: 20–80 µmol/L Tyr: 200–600 µmol/L NTBC: 30–70 µmol/L
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Note: Gender: No difference in tables. NTBC starting dose: 1 mg/kg/d, increase to 2 mg/kg/d in acute severe liver failure. Monitor AFP for hepatocellular carcinoma risk.

Daily Nutrient Requirements by Age

Age	Phe + Tyr (mg/kg/d)	Total Protein (g/kg/d)	Energy (kcal/kg/d)
0–3 mo	65–155	3.0–3.5	95–145 (mean 120)
3–6 mo	55–135	3.0–3.5	95–145 (mean 120)
6–9 mo	50–120	2.5–3.0	80–135 (mean 110)
9–12 mo	40–105	2.5–3.0	80–135 (mean 105)
1–4 yr	380–800 mg/day	≥30 g/day	900–1800 (mean 1300)

Reference: Appendix K-6 | Chinsky et al. Genet Med 2017 | Acosta 2001

3. Homocystinuria (HCU — CBS Deficiency)

Deficient Enzyme Cystathionine beta-synthase (CBS)	Toxic Metabolite Homocysteine (tHcy)	Treatment Goals tHcy <50 µmol/L (B6-responsive) <100 µmol/L (non-responsive) Met <1000 µmol/L Cys: normal
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Note: Gender: No difference in main tables. During pregnancy: consult metabolic team. Conversion: 20 mg Met ≈ 1 g protein.

Daily Nutrient Requirements by Age

Age	Methionine (mg/kg/d)	Cystine (mg/day)	Intact Protein	Total Protein (g/kg/d)
0–6 mo	15–60	85–150	60–100% DRI	100–140% DRI
6–12 mo	12–43	85–150	60–100% DRI	100–140% DRI
1–4 yr	9–28	60–100	60–100% DRI	100–140% DRI
4–7 yr	7–22	50–80	60–100% DRI	100–140% DRI
7–11 yr	7–22	30–50	60–100% DRI	100–140% DRI
>11 yr	See Chapter 15	—	Based on tHcy & Met	100–140% DRI

Vitamin & Supplement Therapy

Supplement	Dose	Purpose
Pyridoxine (B6)	100 mg/day (test dose)	Assess B6-responsiveness (>50% drop in tHcy = responsive)
Folate	5–10 mg/day (or folic acid 1–5 mg/day)	Correct deficiency before B6 assessment
Vitamin B12	Dose based on deficiency	Correct deficiency
Betaine (Cystadane)	Children: 50 mg/kg BID Adults: 3 g BID	For non-B6-responsive patients

Reference: Appendix K-2 | Morris et al. J Inherit Metab Dis 2017 | Roberts AM. Springer 2022

4. Maple Syrup Urine Disease (MSUD)

Deficient Enzyme BCKADH complex	Toxic Metabolite Leucine + 2-oxo-isocaproic acid	Treatment Goals Leu: 100–300 Ile: 100–300 Val: 200–400 µmol/L
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Note: Gender: Table applies to males and non-pregnant/non-lactating females. Pregnant/lactating women: different requirements — see Chapter 12. Conversion: 60 mg Leu ≈ 1 g protein.

Acute Decompensation Requirements (Infant 0–12 mo)

Status	Energy (kcal/kg)	Lipid (% energy)	Protein (g/kg)	Leucine	Isoleucine	Valine
Acute	120–140	40–50%	3–4	0 (withhold)	80–120 mg/kg	80–120 mg/kg
Asymptomatic	100–120	DRI	2–3.5	40–100 mg/kg	30–95 mg/kg	30–95 mg/kg

Stable Maintenance Requirements by Age

Age	Leu (mg/kg/d)	Ile (mg/kg/d)	Val (mg/kg/d)	Protein (g/kg/d)	Energy (kcal/kg/d)
0–6 mo	40–100	30–90	40–95	2.5–3.5	95–145
7–12 mo	40–75	30–70	30–80	2.5–3.0	80–135
1–3 yr	40–70	20–70	30–70	1.5–2.5	80–130
4–8 yr	35–65	20–30	30–50	1.3–2.0	50–120
9–13 yr	30–60	20–30	25–40	1.2–1.8	40–90
14–18 yr	15–50	10–30	15–30	1.2–1.8	35–70
≥19 yr (M & F*)	15–50	10–30	15–30	1.1–1.7	35–45

Illness Protocol

Action	Details
Reduce Leu	Reduce intact protein 50–100% depending on plasma Leu and illness severity
Add Ile + Val	20–120 mg/kg/day each Target: 400–800 µmol/L
Restart Leu source	When Leu <200 µmol/L (infants/<5 yr) or <300 µmol/L (>5 yr)

Reference: Table 17.1 + Table 17.2 | van Calcar S. Springer 2022 | SERN-GMDI MSUD Guidelines

5. Urea Cycle Disorders (UCD)

Enzymes NAGS / CPS / OTC / ASS / ASL / Arginase	Toxic Metabolite Ammonia (all UCD) ASA (ASL deficiency) Arginine (Arginase def.)	Treatment Goals Ammonia <35 µmol/L (<60 mcg/dL) Plasma AA: all normal
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Note: Gender: OTC is X-linked — ~15% of female carriers require treatment. Pregnancy: high risk of ammonia crisis — requires immediate metabolic team contact. No difference in the main tables for other UCDs.

Daily Protein Requirements by Age

Age	Intact Protein (g/kg/d)	Medical Food EAA (g/kg/d)	Total Protein (g/kg/d)	Note
0–1 yr	0.8–1.1	0.4–1.1	1.2–2.2	30–50% from EAA medical food
1–7 yr	0.7–0.8	0.3–0.7	1.0–1.2	—
7–19 yr	0.3–1.0	0.4–0.7	0.8–1.4	—
>19 yr	0.6–0.7	0.2–0.5	0.8–1.0	—

Ammonia Levels — Interpretation

Ammonia (µmol/L)	(mcg/dL)	Interpretation
<35	<60	Normal
36–60	60–100	Mild elevation; may be asymptomatic
61–200	150–350	Elevation: poor feeding, vomiting, lethargy, confusion
>250	>350	Hyperammonemic crisis — coma risk

Supplementation by UCD Type

UCD Type	Supplement	Dose
ASS / ASL (Citrullinemia)	L-Arginine	100–300 mg/kg/day (100 mg/kg may suffice in ASL)
OTC / CPS	L-Citrulline	100–200 mg/kg/day
Arginase deficiency	None (no Arg/Cit)	Protein restriction only
All UCD	Nitrogen scavengers	Sodium benzoate + phenylbutyrate/Ravicti — per protocol

Reference: Table 16.3 | Table 16.1 | MacLeod E. Springer 2022

6. Glutaric Acidemia Type 1 (GA-1)

Deficient Enzyme Glutaryl-CoA dehydrogenase (GCDH)	Toxic Metabolites 3-hydroxyglutaric acid Glutaric acid Glutaryl carnitine	Treatment Goals Plasma lysine: low-normal range Free carnitine: normal
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Note: Gender: No difference. After age 6–7 yr: liberalize protein to age-appropriate DRI. No evidence of benefit from continued strict restriction after age 6 yr. Conversion: 35 mg Lys ≈ 1 g protein. L-Carnitine: 100 mg/kg/day always.

Daily Nutrient Requirements by Age

Age	Total Protein (g/kg/d)	Lysine (mg/kg/d)	Tryptophan (mg/kg/d)	L-Carnitine
Birth–6 mo	2.75–3.0	65–100	10–20	100 mg/kg/day
6–12 mo	2.5–3.0	55–90	10–12	100 mg/kg/day
1–4 yr	1.8–2.6	50–80	8–12	100 mg/kg/day
4–7 yr	1.6–2.0	40–70	7–11	100 mg/kg/day
>6–7 yr	DRI (liberalize)	Low-normal range	—	100 mg/kg/day

Plasma Amino Acid Reference Ranges (μmol/mL)

Amino Acid	0–1 month	1–24 months	2–18 yr	Adult
Arginine	6–140	12–133	10–140	15–128
Lysine	92–325	52–196	48–284	100–250
Tryptophan	—	5–60	34–47	42–106

Emergency Protocol (Acute Illness)

Action	Details
Reduce intact protein	50% within 24–36 hours
Increase lysine-free medical food	50% increase
IV Glucose	10% dextrose at 1.5× maintenance fluid + Intralipid 2 g/kg/day
L-Carnitine IV	Continuous infusion during admission

Reference: Table 20.1 | Table 20.3 | Boy et al. *J Inherit Metab Dis* 2017 | Bernstein LE. Springer 2022

7. Propionic Acidemia & Methylmalonic Acidemia (PA / MMA)

PA Enzyme
Propionyl-CoA carboxylase Cofactor: Biotin

MMA Enzyme
Methylmalonyl-CoA mutase Cofactor:
Adenosylcobalamin (B12)

Restricted AAs
Valine, Isoleucine, Methionine, Threonine
(Propiogenic AAs)

Note: Gender: No difference in tables. MMA B12-responsiveness test: 1.0 mg hydroxocobalamin IM/IV x5 days — ≥50% reduction in MMA = responsive. PA: biotin 5–40 mg/day (test only). L-Carnitine: 100–300 mg/kg/day ÷ 2–4 doses.

Daily Nutrient Requirements by Age

Age	Intact Protein (g/kg/d)	Total Protein (g/kg/d)	Energy (kcal/kg/d)
0–3 mo	0.9–1.5	1.5–1.8	72–109
3–6 mo	0.9–1.5	1.5–1.8	72–109
7–12 mo	0.7–1.2	1.2–1.4	64–97
1–3 yr	0.6–1.05	1.0–1.2	66–99
4–8 yr	0.57–0.95	0.95–1.1	56–88

Acute Illness Protocol

Phase	Action
Acute onset	Stop all protein + D10W 1–1.5x maintenance + Lipid 2 g/kg/day IV
After ammonia improves	Start parenteral EAA at 0.5 g/kg/day → advance 0.25 g/kg/day
Transition to enteral	Propiogenic AA-free medical food + intact protein (breast milk or standard formula)

Reference: Appendix K-4 | SERN-GMDI PROP Guidelines | Sowa M. Springer 2022

8. Long-Chain Fatty Acid Oxidation Disorders (VLCAD / LCHAD)

VLCAD Enzyme Very long-chain acyl-CoA dehydrogenase	LCHAD Enzyme Long-chain 3-hydroxyacyl-CoA dehydrogenase	Key Restriction Long-chain fat (LCF) + fasting avoidance
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Note: Gender: No difference. Total fat is NOT restricted — only the SOURCE (LCF). LCHAD specifically requires DHA: <20 kg → 60 mg/day | >20 kg → 100 mg/day. Triheptanoin (Dojolvi): up to 35% of total energy as alternative to even-chain MCT.

Fat Distribution by Age and Disease Severity (VLCAD)

Age	Severity	LCF (% energy)	MCT (% energy)	Total Fat (% energy)
0–6 mo	Severe	10–15	30–45	~40–55
0–6 mo	Moderate	15–30	10–30	~35–45
0–6 mo	Mild	30–55	0–20	~30–55
7–12 mo	Severe	10–15	25–30	~35–42
7–12 mo	Moderate	15–30	10–25	—
7–12 mo	Mild	30–40	0–10	—
1–3 yr	Severe	10–15	10–30	~30–40
1–3 yr	Moderate	20–30	10–20	—
1–3 yr	Mild	20–40	0–10	—
4–18 yr	Severe	—	15–25	10%
4–18 yr	Moderate	—	10–20	15–25%
4–18 yr	Mild	—	0–10	20–35%
>19 yr	Severe	—	10–25	10%
>19 yr	Moderate	—	10–20	15–20%
>19 yr	Mild	—	0–10	20–35%

Maximum Fasting Intervals (Well Patient — VLCAD)

Age	Maximum Hours Between Feeds
0–4 months	3–4 hours
4–6 months	4–6 hours
6–9 months	6–8 hours
9–12 months	8–10 hours
>12 months	10–12 hours

Reference: Table 23.3 | *SERN-GMDI VLCAD Nutrition Guidelines 2019* | Rohr F. Springer 2022

9. MCAD Deficiency (Medium-Chain Acyl-CoA Dehydrogenase)

Deficient Enzyme Medium-chain acyl-CoA dehydrogenase (MCAD)	Key Restriction Fasting only — no fat restriction	Treatment Goal Prevent hypoketotic hypoglycemia Carnitine: normal
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Note: Gender: No difference. Most common FAOD. Do NOT give MCT (system cannot oxidize medium-chain FA). Protein, carbohydrates, fat: normal DRI. Key = fasting avoidance. Supplement L-carnitine only if plasma levels are low.

Maximum Fasting Intervals

Age	Maximum Hours Between Feeds
0–4 months	3–4 hours
4–6 months	4–6 hours
6–9 months	6–8 hours
9–12 months	8–10 hours
>12 months	10–12 hours

Illness Protocol

Condition	Action
Well — no illness	Normal unrestricted diet Follow fasting guidelines
Mild illness	Increase carbohydrates + frequent feeding + avoid fasting
Severe illness / poor intake	IV D10W at maintenance rate — do NOT allow fasting

Reference: SERN-GMDI FAOD Guidelines | Chapter 23 | Merritt et al. GeneReviews

10. Classic Galactosemia

Deficient Enzyme GALT — Galactose-1-phosphate uridylyltransferase	Toxic Metabolites Galactose-1-phosphate Galactitol	Treatment Goal RBC Gal-1-P <1 mg/dL (or lowest achievable)
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Note: Gender: Females — monitor ovarian function (FSH/LH/E2) from puberty due to risk of primary ovarian insufficiency (POI). Both sexes: calcium + vitamin D supplementation critical due to dairy exclusion. No specific nutrient requirements differ by age — lifelong galactose restriction is the sole dietary intervention.

Allowed and Restricted Foods (Lifelong)

Category	Foods / Ingredients
ALLOWED — Galactose-free	Soy-based infant formulas (soy protein isolate), amino acid-based formulas All fruits, vegetables and their juices All legumes (beans, soybeans) Non-fermented soy products (soy milk, tofu, TVP) Aged cheeses (<25 mg Gal/100g): Jarlsberg, Emmentaler, Swiss, Gruyere, sharp cheddar, Parmesan (>10 mo) Sodium/calcium caseinate All cocoa except milk chocolate
ALLOWED in moderation	Organ meats (galactose content unknown — no direct evidence of harm) Fermented soy products (miso, natto, tempeh) — small condiment amounts only
RESTRICTED — lifelong	Breast milk All milk-based infant formulas All dairy products (milk, yogurt, soft cheese, ice cream) Lactose in any ingredient label Most cheeses (except aged cheeses listed above) Beano-type products (release galactose from plant foods)

Nutritional Monitoring

Nutrient / Test	Requirement / Target
Calcium	Age-appropriate DRI — especially important without dairy
Vitamin D (25-OH)	Target 20–32 ng/mL (50–80 nmol/L) — annual assessment
Bone density (DXA)	First scan at age 8–10 yr z-score ≤ -2 → follow-up Repeat after puberty
Ovarian function (Females)	FSH, LH, E2 from puberty annually
RBC Gal-1-Phosphate	Periodic monitoring in first year; annually thereafter

Reference: Table 24.1 | Welling et al. *J Inherit Metab Dis* 2017 | Bernstein LE & van Calcar S. Springer 2022

11. Glycogen Storage Disease Type I — Von Gierke (GSD-I)

GSD Ia Enzyme Glucose-6-phosphatase	GSD Ib Enzyme G6P transporter (+ neutropenia → G-CSF)	Key Restriction Fructose + galactose + fasting Target glucose: 70–120 mg/dL
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Note: Gender: No difference. 1 tablespoon UCCS = 8 g = 7.2 g carbohydrate. GSD Ib: additional G-CSF for neutropenia. Cornstarch must NOT be heated, cooked, or mixed with acidic beverages. Diet macros: 65–70% carbohydrate, ~15% fat, normal protein DRI.

Glucose Infusion Rate (GIR) Requirements by Age

Age	Required GIR (mg/kg/min)	Glucose Source
0–12 months	7–9	Continuous NG drip overnight
1–3 years	7	Uncooked cornstarch (UCCS)
3–6 years	6–7	UCCS
6–14 years	5–6	UCCS
Adolescent	4–5	UCCS
Adult	3–4	UCCS

Glycosade Dosing by Age (Overnight Use)

Age	Glycosade Dose (g)
5–6 years	60–75
7–8 years	75–90
Pre-pubertal	90–120
Pubertal	135–150
Adult	120–150

Hypoglycemia Emergency Protocol

Condition	Action
Asymptomatic BG <70 mg/dL	5 g carbohydrate from food (pretzels/crackers) → recheck in 10 min
Symptomatic hypoglycemia	10 g dextrose (glucose gel/tablets) + food or 1 hr UCCS dose → recheck
Illness / hospital	IV D10W–D12.5W at GIR rate via peripheral line + low threshold for admission

Reference: Appendix I | Fernandes et al. 2000 | Sowa M. Springer 2022 | SERN-GMDI

12. Glycogen Storage Disease Type III — Cori/Forbes (GSD-III)

Deficient Enzyme
AGL — Debranching enzyme
(amylo-1,6-glucosidase)

Type IIIa
Liver + Muscle (85% of cases)

Type IIIb
Liver only (15%)

Note: Gender: No difference. Unlike GSD-I: fructose and galactose are NOT restricted (gluconeogenesis intact). Alcohol impairs gluconeogenesis — must be avoided. High protein intake is the cornerstone of treatment for muscle protection.

Macronutrient Distribution

Macronutrient	Target (% of Total Energy)	Amount (g/kg/day)	Notes
Carbohydrate	35–55%	—	Complex carbohydrates preferred; limit simple sugars
Protein	20–30%	3–4 g/kg/day	High protein supports gluconeogenesis and muscle function
Fat	20–35%	—	No specific restriction

UCCS / Glycosade Dosing for GSD-III

Timing	Dose / Target
UCCS (if needed)	1 g/kg every 4 hours (less than GSD-I requirements)
High-protein bedtime snack	Protein + UCCS or Glycosade to extend overnight fasting
Glycosade (ketotic forms)	30–75 g at bedtime

Reference: GSD III Guidelines 2010 | Chapter 26 | Sowa M. Springer 2022

LABORATORY MONITORING — QUICK REFERENCE

Disease	Frequency	Key Labs
PKU	Weekly (infancy), weekly–monthly thereafter	Blood Phe, Tyrosine, Plasma AA, Prealbumin, 25-OH Vit D, CBC, Zn, Cu, B12, EFA
HT-1	Monthly (yr 1), q3mo (yr 1–5), q6mo (yr 5–10), annually after	Plasma AA, PT/PTT, NTBC level, ALT/AST, CBC, BUN/Cr, Ca, P
HCU	Weekly (infancy), weekly–monthly thereafter	thCys total, Plasma Met, Plasma AA, B12, Folate, EFA, 25-OH Vit D, Zn, Se
MSUD	Daily until stable, weekly to 6 mo, monthly after, 24-hr urine after	Plasma AA, Ketones, Prealbumin, Albumin, Ferritin, CBC
UCD	Weekly (infancy), monthly thereafter	Ammonia, Plasma AA (esp. glutamine), Prealbumin, 25-OH Vit D, CBC, Fe, Folate, Zn
GA-1	q3mo to 1 yr; q6mo to 6 yr; annually after	Plasma AA (Lys, Arg, Trp), Carnitine (free+acyl), Albumin, CBC, Ca, P, Ferritin, B12
PA/MMA	Monthly (infancy); q3–6mo after	Plasma AA, Serum MMA, Carnitine, Ketones, CBC, Albumin, Prealbumin, Organic acids
LC-FAOD	q3mo (Cr. kinase, carnitine, acylcarnitine)	Plasma AA, Carnitine, Acylcarnitine, EFA, BNP (LCHAD), DHA levels (LCHAD), CBC
Galactosemia	q3–6mo yr 1; annually after	RBC Gal-1-P, 25-OH Vit D, DXA (age 8–10 yr), FSH/LH/E2 (females at puberty)
GSD-I	q3–6mo	Glucose (home monitor), Lactate, Uric acid, TG, Cholesterol, LFT, AFP + liver US (annual)
GSD-III	q3–6mo	Glucose (home monitor), CK, LFT, Albumin, Prealbumin, Echocardiogram (periodic)

DISCLAIMER: This reference document is extracted for educational purposes from Nutrition Management of Inherited Metabolic Diseases, 2nd Edition (Springer 2022), edited by Bernstein, Rohr & van Calcar. All clinical decisions must be made in consultation with a qualified metabolic team. Requirements may vary based on individual disease severity, genotype, growth status, and biochemical response.